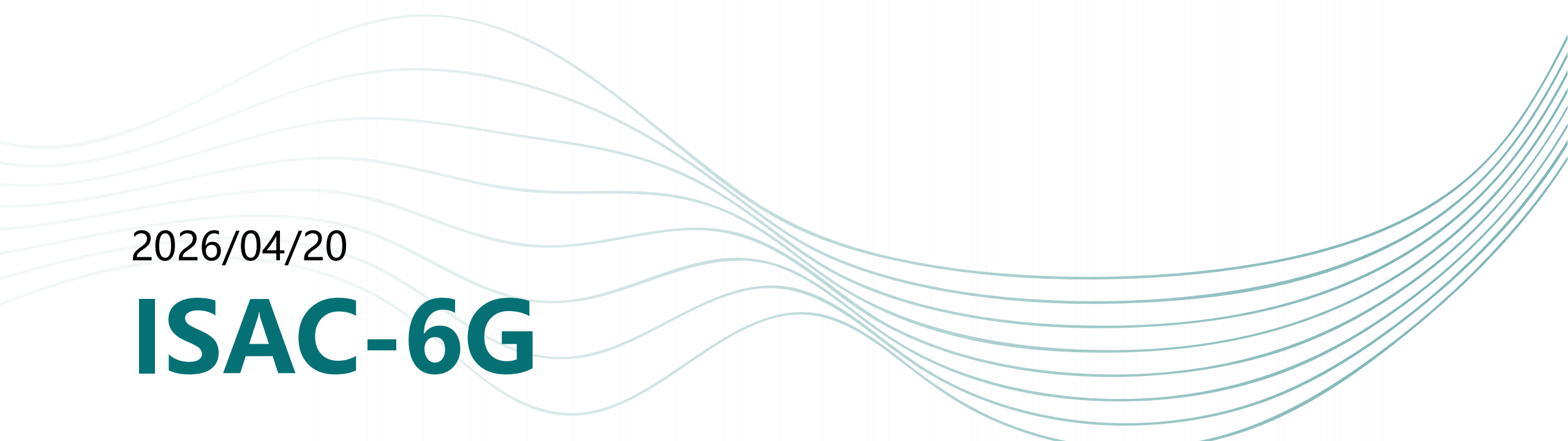
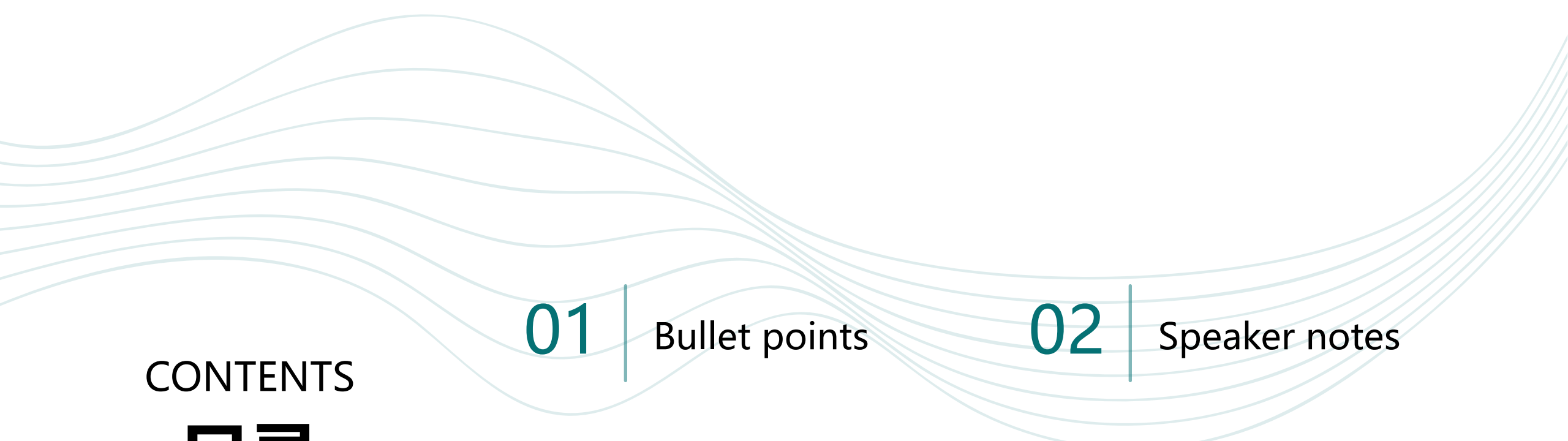


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2026/04/20

ISAC-6G

汇报人: Muhammad
Umar Farooq Qaisar



CONTENTS

目录

01

| Bullet points

02

| Speaker notes

03

| Core Challenges
Summary

What is ISAC for 6G



ISAC Definition

Deep integration of communication and sensing via unified signal processing, hardware, and waveforms.



ISAC Advantages

Improves spectral efficiency, reduces cost/energy use, enables real-time environmental intelligence.

What is ISAC for 6G



6G Role

Evolves to dual-function joint transceivers, cornerstone of adaptive intelligent systems.

Key Technologies

RIS enhances coverage, AI optimizes performance, novel waveforms improve accuracy/robustness.

An abstract graphic consisting of multiple thin, teal-colored lines. These lines originate from the top edge of the frame and curve downwards and to the right, eventually converging into a single point on the right side of the image. From this convergence point, several lines curve downwards and to the left, creating a fan-like or tail-like shape that extends towards the bottom right corner.

01

Bullet points

Standardization



Leading Organizations

3GPP Rel- 19/20, IEEE 802.11bf, ETSI ISG lead standardization efforts.



Key Responsibilities

Define architectures and KPIs, details on pages 20–22.

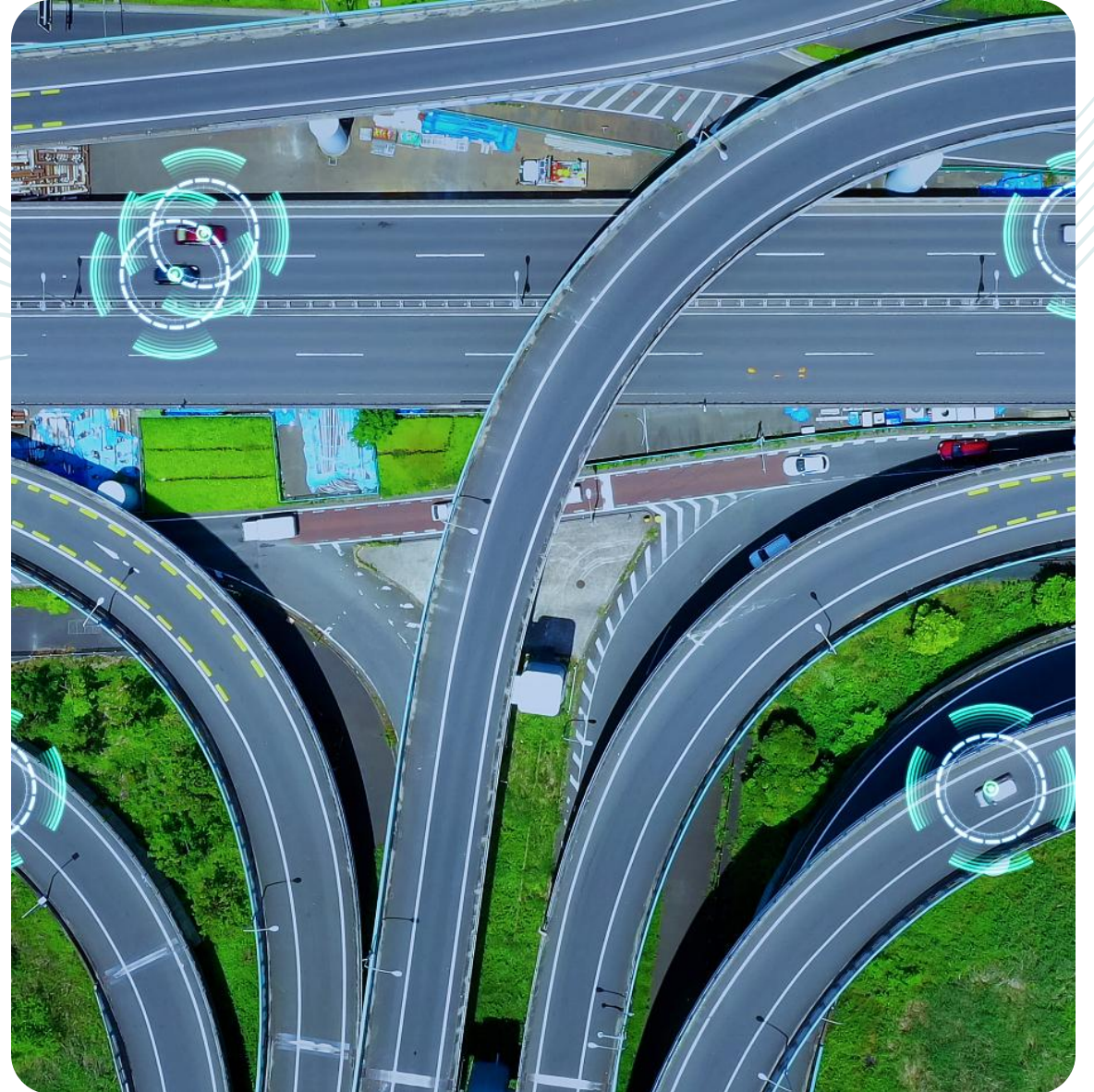
Applications

Intelligent transportation

Cm-level positioning for intelligent transportation applications.

Industrial IoT

Industrial IoT as an application scenario mentioned.



Applications



Immersive XR

Immersive XR included in application scenarios.

Info pages

Relevant info on pages 21 and 26 for applications.

Testbeds



Testbeds content

Involves mmWave prototyping platforms, driving algorithm validation, covered in pages 23–24.



Datasets content

Includes multi-modal datasets, aiding algorithm validation, covered in pages 23–24.



02

Speaker notes

Base Station



Industry Leaders

Companies like Huawei and Nokia are leading base station transformation efforts.



Functional Transformation

Base stations are being turned into multi-function sensing-communication hubs.

RIS Sensing



RIS Sensing Improvement

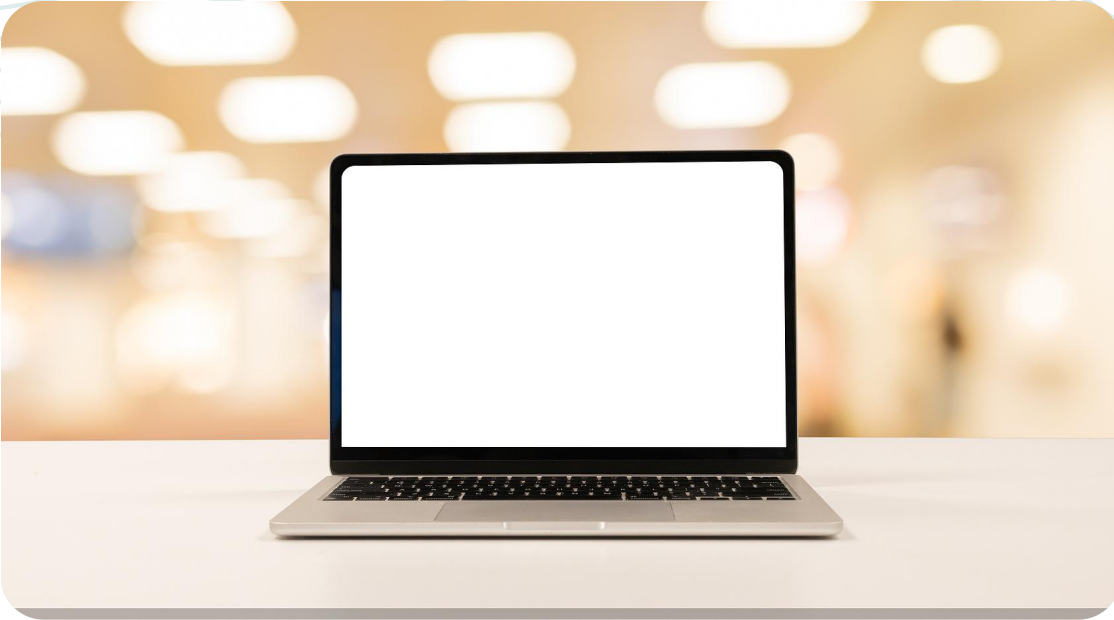
Testbeds indicate RIS enhances sensing fidelity in complex environments (page 23).



RIS Sensing Bottleneck

Lack of unified evaluation standards remains a key issue (page 24).

ISAC Application



Application Area

Autonomous driving applications require ISAC for cooperative sensing.



Performance Requirement

Achieves cooperative sensing with millisecond-level latency.



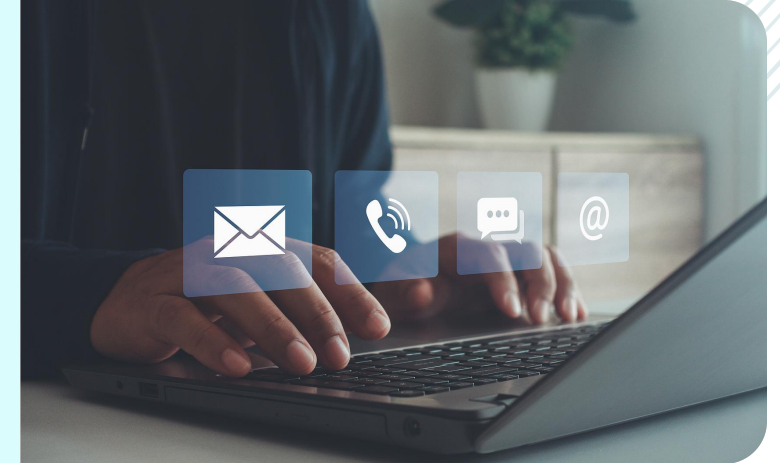
03

Core Challenges Summary

Slide Content: ISAC Obstacles

Interference

Sensing and communication share resources, causing mutual interference needing dynamic beamforming, scheduling, and AI-adjusted real-time waveforms.



Hardware

Non-linear effects and phase noise in mmWave/THz bands limit performance, and full-duplex designs must address self-interference.

Slide Content: ISAC Obstacles



Protocol Issue

Missing cross-layer protocols hinder multi-vendor interoperability, highest priority per Table VIII.



Privacy Problem

Privacy frameworks lacking, prominent in human-centric sensing applications (pages 25,28).

Slide Content: ISAC Obstacles



KPI Unification

3GPP needs to unify sensing (e.g., detection probability) and communication KPIs (e.g., throughput).



Slide Content: Future ISAC



Evolution Phases

Shifts from data transmission to task-relevant info extraction, evolving in three phases.



Focus Areas

Currently focused on constrained scenarios like industrial robots, long-term goal to quantify task-critical info with "semantic entropy".



Slide Content: Future ISAC

Quantum Integration

Involves quantum sensing improving accuracy and quantum security enhancing communication.

Current Status

At TRL 1 - 2, facing challenges like cryogenic hardware, expected to mature after 6G.

Slide Content: Future ISAC

Security Defense

Cross-layer mechanisms against eavesdropping and tampering for security.

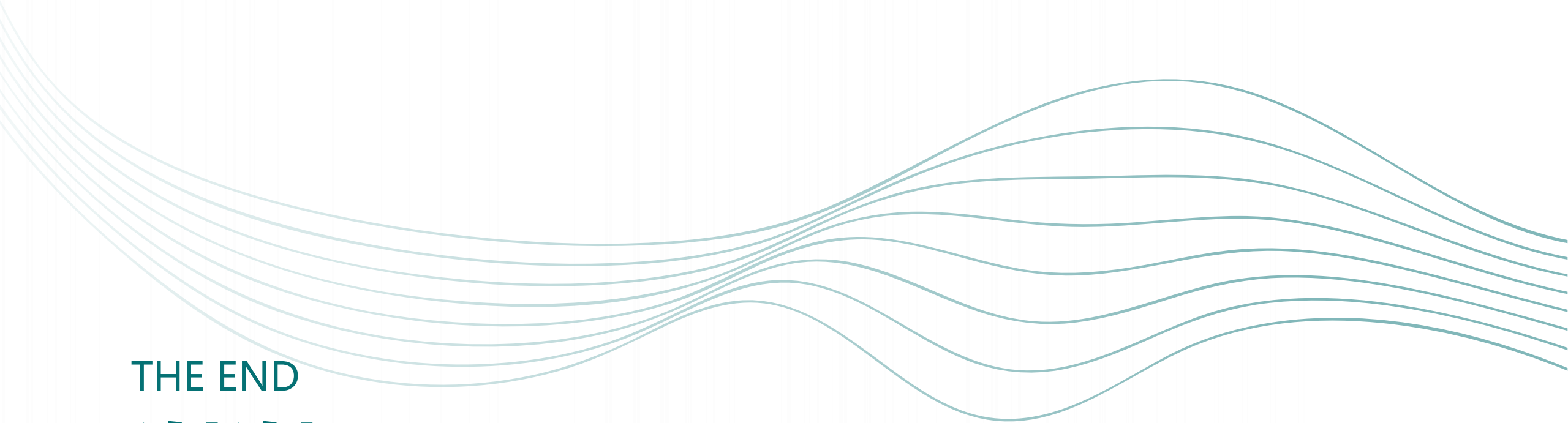
Spectrum Regulation

Dynamic spectrum-sharing regulations need improvement for future ISAC.

Algorithm Design

Anti-jamming algorithms and privacy-preserving sensing protocols must be designed.



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THE END

谢谢